What Drives National Happiness?

Cross-country econometric analysis 132 countries

Research Question

What is the effect of GDP per capita on national happiness, holding healthy life expectancy, Gini, and unemployment constant, across 132 countries in 2025?

Framing: Associational headline with one documented IV stress-test for causal inference.
Data: World Bank WDI, Inflation, Latitudes, WHR (world happiness report)
Workflow: 8-phase framework - research question → robustness, with diagnostic gates between phases.

PHASE 2 — THEORY

Four Regressors, Four Channels, Pre-Committed Signs

log_gdp_pc → happiness (+)

Measures a country's income level after adjusting for purchasing power. Higher income generally improves living standards and access to resources.

life_exp → happiness (+)

Measures average life expectancy at birth. Higher values indicate better health, healthcare quality, and overall living conditions.

gini → happiness (−)

Measures income inequality within a country. Higher inequality can reduce social cohesion and increase feelings of unfairness

unemployment → happiness (−)

Measures the percentage of the labor force without a job. Higher unemployment is associated with financial stress and lower life satisfaction.

Pre-commitment: β_1 on log_gdp_pc is THE headline coefficient. Phase 5 tracks it as controls are added; Phase 7 instruments only it; Phase 8 leads interpretation with it. Prevents post-hoc cherry-picking.

PHASES 3+4 — DATA & MODEL

Empirical Model & Sample

Population Regression Function (PRF)

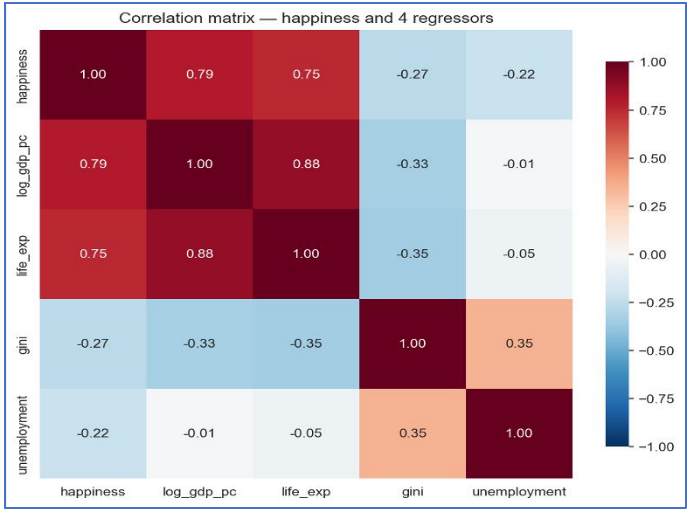
$$happiness\_i = \beta_0 + \beta_1 \, log\_gdp\_pc_i + \beta_2 \, life\_exp_i + \beta_3 \, gini_i + \beta_4 \, unemployment_i + u_i$$

Functional form: log on GDP (constant-elasticity + diminishing returns). Levels on others (no theoretical non-linearity in observed range). Linear in parameters — CLRM A1 satisfied.

Descriptive Statistics (132 countries, zero missing)

Variable	Mean	SD	Min	Max
happiness (0–10)	5.66	1.09	3.25	7.76
log_gdp_pc	9.67	1.11	7.31	11.76
life_exp (years)	73.9	7.18	54.6	84.4
gini (0–100)	36.1	7.28	23.8	59.1
unemployment (%)	6.70	5.60	0.39	34.2

Correlation Matrix



Univariate Correlations with Happiness

GDP **+0.80** · life_exp **+0.75** · gini **−0.27** · unemp **−0.22** — all four signs match priors.

⚠ **Multicollinearity Flag:** ρ(log_gdp_pc, life_exp) = **0.88** → watch for inflated SEs in Phase 5; check VIF in Phase 6.
Sample-size gate: 132 obs / 5 parameters = 26 per coefficient ✓ (rule of thumb > 20–30).

PHASE 5 — ESTIMATION

Baseline Estimation: Full Model M4

Coefficient Table (classical SEs, validated by Phase 6 BP/White)

Variable	$\hat{\beta}$	SE	t	p	95% CI
log_gdp_pc	**+0.630**	0.103	+6.13	<0.001	[+0.43, +0.83]
life_exp	+0.032	0.016	+2.04	0.043	[+0.001, +0.064]
gini	+0.015	0.008	+1.78	0.077	[−0.002, +0.032]
unemployment	−0.047	0.010	−4.52	<0.001	[−0.067, −0.026]

$N = 132 \cdot R^2 = 0.693 \cdot \text{Adj } R^2 = 0.684$

Individual t-tests

$H_0: \beta_j = 0 \quad H_1: \beta_j \neq 0$

GDP and unemp reject at 1%; life_exp at 5%; gini at 10% only.

⚠ **Anomaly Flagged:** Gini coefficient is *positive*, intuitively it should have negative correlation, but p-value not significant

Joint F-test (overall significance)

$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0 \quad H_1$: at least one $\neq 0$

$F(4,127) = 71.80$, $p \approx 10^{-31}$ → reject decisively.

Restriction F-test

H_0: β_gini = β_unemp = 0 $\quad H_1$: at least one $\neq 0$

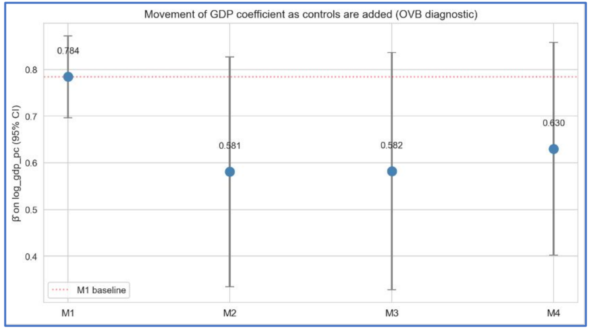
$F(2,127) = 10.23$, $p = 0.0001$ → reject.

PHASE 5 — OVB DIAGNOSTIC

Nested Regressions Reveal Omitted-Variable Bias

Add controls one block at a time, track $\hat{\beta}$_GDP

Spec	Controls added	$\hat{\beta}$_GDP	R^2	Δ from M1
M1	(univariate)	0.784	0.632	—
M2	+ life_exp	**0.581**	0.644	**−0.203 (−26%)**
M3	+ gini	0.582	0.644	−0.20
M4	+ unemployment	0.630	0.693	−0.155



The 26% drop M1 → M2 is the Session-3 OVB formula in action:
$E[b_2] = \beta_2 + \beta_3 \cdot Cov(X_2, X_3) / Var(X_2)$

life_exp positively affects happiness AND correlates with GDP at 0.88 → omitting it lets GDP "mimic" life expectancy → $\hat{\beta}$_GDP inflated upward by ~0.20 happiness points.

Logical implication: If *observable* controls cause this much OVB on β_GDP, *unobservable* ones (social support, freedom, corruption) probably cause more. → Direct motivation for Phase 7 IV attempt and Phase 8 robustness with WHR confounders.

PHASE 6 — CLRM DIAGNOSTICS

Four Formal Tests of OLS Assumptions

VIF - Multicollinearity

GDP = 4.49, life_exp = 4.51, gini = 1.30, unemp = 1.15

All < 5 → **benign** despite ρ(GDP, life_exp) = 0.88

Breusch-Pagan Homoscedasticity Test

H_0: homoskedastic errors H_1: heteroskedastic

LM = 9.09, p = 0.059 → **fail to reject at 5%**

White's Test Homoskedasticity Test

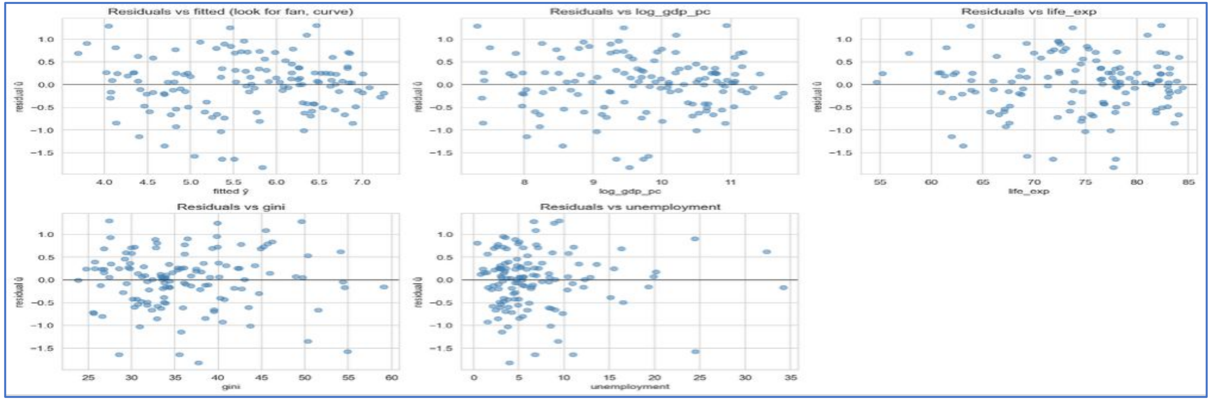
H_0: homoskedastic errors H_1: heteroskedastic

LM = 17.26, p = 0.243 → **fail to reject**

Ramsey RESET Functional Form

H_0: no omitted higher-order terms H_1: model misspecified

Power = 2 ($\hat{y}^2$): F = 0.75, p = 0.39 → fail to reject



Exit verdict (Path ii — patch & proceed):

A4 holds → classical SEs validated. not true non-linearity → addressed by Phase 6.

A2 (endogeneity) is not testable with observational data alone → deferred to Phase 7.

PHASE 7, PART 1 — IV SETUP

Just-Identified IV — |latitude| Passes Relevance

With $k_1 = 1$ (log_gdp_pc only endogenous) and $L = 1$ (|latitude| as excluded instrument), the model is **just-identified ($L = k_1 = 1$)** → IV is feasible. We lose the Sargan over-identification test, acknowledged as a limitation.

Why |latitude| (Distance from Equator)?

Classic geography-and-growth instrument (AJR 2001; Sachs 2003; Diamond 1997). Tropical countries (low |lat|) tend to be poorer due to climate, disease burden, agriculture, and colonial history. Absolute value used because tropical Brazil and tropical Indonesia are both poor regardless of hemisphere.

why only log_GDP is endogenous?
We treat only log GDP per capita as endogenous because the reverse-causality channel between happiness and GDP is well documented (Oswald, De Neve, Stevenson-Wolfers), whereas country-aggregate reverse-causality channels for life expectancy, inequality, and unemployment are much weaker. With one endogenous regressor and one excluded instrument, the model is just-identified and IV is feasible — though we lose the Sargan over-identification test, which we acknowledge as a limitation.

First-Stage Regression

$$\text{log\_gdp\_pc}_i = \pi_0 + \pi_1 |\text{latitude}|_i + \pi_2 \text{ life\_exp}_i + \pi_3 \text{ gini}_i + \pi_4 \text{ unemployment}_i + v_i$$

$H_0: \pi|lat| = 0 \quad H_1: \pi|lat| \neq 0$

First-Stage Coefficient Table

Variable	$\hat{\pi}$	SE	t	p
|latitude|	**+0.0165**	0.0039	**+4.22**	**<0.001**
life_exp	+0.114	0.008	+14.45	<0.001
gini	+0.011	0.008	+1.39	0.168
unemployment	−0.003	0.009	−0.38	0.704

$R^2 = 0.805$

✅ **First-Stage F on excluded instrument = 17.85**
Test Statistic: F > 10 → relevant. Distance from equator has strong independent partial correlation with log GDP per capita after conditioning on the three controls.

PHASE 7, PART 2 — 2SLS + EXCLUSION

2SLS Works Mechanically

OLS vs. 2SLS on $\hat{\beta}$_GDP

	OLS (Ph.5)	2SLS (Ph.7)
$\hat{\beta}$GDP	+0.630	**+1.168**
SE	0.103	0.316
p-value	<0.001	**0.0002**
95% CI	[+0.43, +0.83]	**[+0.55, +1.79]**

→ 2SLS estimate is **85% larger** than OLS. SEs reasonable; coefficients interpretable. No weak-IV blow-up.

IV Coefficient Table

$$happiness\_i = \alpha + \beta_1^{IV} \widehat{\log}(GDPpc\_i) + \beta_2\ life\_exp\_i + \beta_3\ gini\_i + \beta_4\ unemp\_i + e\_i$$

Variable	$\hat{\pi}$	SE	t	p
Intercept	-3.0647	0.8028	-3.8176	<0.001
$\widehat{\log}$(GDPpc_i)	**1.1677**	**0.3165**	**3.6899**	**0.0002**
life_exp	-0.0394	0.0431	-0.9143	0.3606
unemployment	-0.0520	0.0115	-4.5095	0.0000
gini	−0.003	0.009	−0.38	0.704

Durbin-Wu-Hausman Test

H_0: OLS consistent (GDP exogenous) H_1: OLS biased

⚠ Coef on $\hat{v}$ = −0.61, t = −1.99, **p = 0.049 → reject at 5%**
Under the assumption that |latitude| is valid, OLS is biased and IV is preferred.

Combined IV Verdict

Criterion	Result
Relevance (F > 10)	F = 17.85

→ Causal effect not point-identified. Bracketed result on next slide.

PHASE 7 — HEADLINE

Bracketing the Causal Effect of GDP on Happiness

+0.630

OLS $\hat{\beta}_{GDP}$ (Phase 5 M4)

Partial association, conditional on the 3 controls.

+1.168

2SLS $\hat{\beta}_{GDP}$ (with |lat| IV)

Causal estimate *if* |latitude| is a valid instrument.

0.6–1.2

The Defensible Bracket

Range covering both biased estimators — the honest bound.

Why a Bracket, Not a Point Estimate

Biases pulling OLS DOWN	Biases pulling OLS UP	Biases pulling 2SLS UP	Biases pulling 2SLS DOWN
Attenuation from measurement error in GDP per capita (national accounts noisy, especially for developing countries)	Reverse causality: happy populations are more productive	|latitude| has direct effects on happiness — climate, institutions, culture — which flow into the 2SLS estimator through exclusion failure	Weak-instrument bias toward OLS — not relevant here (F = 17.85, well above threshold)
	OVB from omitted soft variables (D2.3): social support, freedom, corruption		

★ Defensible Statement ★

The true causal effect of log GDP per capita on happiness is **positive and lies between roughly 0.6 and 1.2 happiness points per unit of log GDP per capita.** Both estimators are biased in unknown net amounts, so we cannot pin a point estimate without a second valid instrument.

★ Bottom Line ★

GDP raises happiness. Unemployment lowers happiness.
Both directions are robust. The magnitudes are bracketed but cannot be pinned without better instruments.

This is what an honest cross-section econometric study of happiness looks like - clear signs, clear orders of magnitude, defensible bounds, and explicit acknowledgement of what cannot be claimed.